

Linear Mixed Model (선형혼합모형)

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Today's Topics

Linear Mixed Model

- Preliminary Model
- Parameter Estimation
- Analysis with LMM
- Marginal and Random Effect Model

Preliminary Model

Linear Model

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \mathbf{Z} \mathbf{b} + \boldsymbol{\varepsilon},$$
$$\mathbf{b} \sim \text{MVN}(\mathbf{0}, \mathbf{G}), \boldsymbol{\varepsilon} \sim \text{MVN}(\mathbf{0}, \mathbf{R}), \text{ indep}$$

The implied marginal model equals

$$\mathbf{Y} \sim \text{MVN}(\mathbf{X}\boldsymbol{\beta}, \mathbf{ZGZ}^t + \mathbf{R})$$

Note that inference based on the marginal model do not explicitly assume the presence of random effects representing the natural heterogeneity between subjects

Preliminary Model (Cont'd)

Notation

β : vector of fixed effects

α : vector of all variance components in $\mathbf{G}$ and $\mathbf{R}$

$\theta = (\beta^t, \alpha^t)^t$: vector of all parameters in marginal model

Marginal likelihood function:

$$L_{ML}(\theta) = \prod_{i=1}^N \left\{ (2\pi)^{-n_i/2} |V_i(\alpha)|^{-\frac{1}{2}} \times \exp\left(-\frac{1}{2}(Y_i - X_i\beta)^t V_i^{-1}(\alpha)(Y_i - X_i\beta)\right) \right\}$$

If α were known, MLE of β equals

$$\hat{\beta}(\alpha) = \left(\sum_{i=1}^N X_i^t W_i X_i \right)^{-1} \sum_{i=1}^N X_i^t W_i y_i$$

where W_i equals V_i^{-1}

Preliminary Model (Cont'd)

In most cases, α is not known, and needs to be replaced by an estimate $\hat{\alpha}$

Two frequently used estimation methods for α :

- Maximum likelihood

- Restricted maximum likelihood

Parameter Estimation

Maximum likelihood estimation (MLE)

$\hat{\alpha}$ is obtained from maximizing $L_{ML}(\alpha, \hat{\beta}(\alpha))$ with respect to α

The resulting estimate $\hat{\beta}(\hat{\alpha}_{ML})$ for β will be denoted by $\hat{\beta}_{ML}$

$\hat{\beta}_{ML}$ and $\hat{\alpha}_{ML}$ can also be obtained from maximizing $L_{ML}(\theta)$ with respect to θ , for example, with respect to α and β simultaneously

Parameter Estimation (Cont'd)

Restricted maximum likelihood estimation (REML)

Consider a sample of N observations $Y_1, \dots, Y_N$ from $N(\mu, \sigma^2)$

For known μ , MLE of σ^2 equals to $\hat{\sigma}^2 = \sum_i (Y_i - \mu)^2 / N$.

Then, $E(\hat{\sigma}^2) = \sigma^2$: unbiased estimator

When μ is unknown, MLE of σ^2 equals to $\hat{\sigma}^2 = \sum_i (Y_i - \bar{Y})^2 / N$

Then, $E(\hat{\sigma}^2) = \frac{N-1}{N} \sigma^2$, $\hat{\sigma}^{2UE} = \sum_i (Y_i - \bar{Y})^2 / (N-1)$

Apparently, having to estimate μ introduces bias in MLE of σ^2

Parameter Estimation (Cont'd)

How can we estimate σ^2 without estimating μ ?

The model for all data simultaneously:

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_N \end{pmatrix} \sim N \left(\begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix}, \sigma^2 I_N \right)$$

We transform Y such that μ vanishes from the likelihood:

$$U = \begin{pmatrix} Y_1 - Y_2 \\ Y_2 - Y_3 \\ \vdots \\ Y_{N-2} - Y_{N-1} \\ Y_{N-1} - Y_N \end{pmatrix} = A^T Y \sim N(0, \sigma^2 A^T A)$$


Independent with μ !!!

Parameter Estimation (Cont'd)

$$\log L(\sigma^2) \propto -\frac{N-1}{2} \log \sigma^2 - \frac{1}{2\sigma^2} Y^t A (A^t A)^{-1} A^t Y$$

$$\frac{\partial}{\partial \sigma^2} \log L(\sigma^2) = -\frac{N-1}{2\sigma^2} + \frac{1}{2\sigma^4} Y^t A (A^t A)^{-1} A^t Y$$

$$\Rightarrow \hat{\sigma}^2 = \frac{1}{N-1} Y^t A (A^t A)^{-1} A^t Y$$

$$E(\hat{\sigma}^2) = \frac{1}{N-1} E\left[Y^t A (A^t A)^{-1} A^t Y\right] = \frac{1}{N-1} E\left[\text{tr}\left\{Y^t A (A^t A)^{-1} A^t Y\right\}\right]$$

$$= \frac{1}{N-1} \text{tr}\left[A (A^t A)^{-1} A^t E\{YY^t\}\right] = \sigma^2$$

Parameter Estimation (Cont'd)

REML

MLE of σ^2 , based on $\mathbf{U}$, equals

$$S^2 = \frac{1}{N-1} \sum_i (Y_i - \bar{Y})^2$$

$\mathbf{A}$ defines a set of $N - 1$ linearly independent 'error contrasts'

S^2 is called the REML estimate of σ^2

S^2 is independent of $\mathbf{A}^t \mathbf{Y}$

Parameter Estimation (Cont'd)

REML

Consider a sample of N observations $Y_1, \dots, Y_N$ from linear regression model $N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$

MLE of σ^2 : $\hat{\sigma}^2 = (Y - X\hat{\boldsymbol{\beta}})^t(Y - X\hat{\boldsymbol{\beta}}) / N$

Note that $\hat{\sigma}^2$ is biased for σ^2 :

$$\begin{aligned} E(\hat{\sigma}^2) &= \frac{1}{N} E[(Y - X\hat{\boldsymbol{\beta}})^t(Y - X\hat{\boldsymbol{\beta}})] = \frac{1}{N} E[Y(I - X(X^t X)^{-1} X^t)Y] \\ &= \frac{1}{N} \text{tr} \left[E \left\{ (I - X(X^t X)^{-1} X^t) Y Y^t \right\} \right] = \frac{\sigma^2}{N} \text{tr} [I - X(X^t X)^{-1} X^t] \\ &= \frac{N-p}{N} \sigma^2 \end{aligned}$$

Parameter Estimation (Cont'd)

REML

Unbiased estimate for σ^2 :

$$\text{MSE} = (Y - X\hat{\beta})^t (Y - X\hat{\beta}) / (N - p)$$

The MSE can also be obtained from transforming the data orthogonal to X :

$A^t Y$: error contrast (A : $N \times (N - p)$)

We consider the error contrast A which satisfies $A^t A = I$ and $AA^t = I - X(X^t X)^{-1}X^t$ for computation:

$$A^t Y \sim N(0, \sigma^2 A^t A)$$

The MSE is again called the REML estimate of σ^2

Parameter Estimation (Cont'd)

REML

Existence of $U = A$:

$I - X(X^t X)^{-1} X^t$: symmetric and idempotent matrix

The eigenvalues of the idempotent and symmetric matrix are always 0 or 1. Thus, by spectral decomposition,

$$1) I - X(X^t X)^{-1} X^t = (U \quad L) \begin{pmatrix} I_{(n-p) \times (n-p)} & \mathbf{0}_{(n-p) \times p} \\ \mathbf{0}_{p \times (n-p)} & \mathbf{0}_{p \times p} \end{pmatrix} \begin{pmatrix} U^t \\ L^t \end{pmatrix}$$

$$2) \begin{pmatrix} U^t \\ L^t \end{pmatrix} (U \quad L) = I_{n \times n}$$

$$\Rightarrow \text{(i): } UU^t = I - X(X^t X)^{-1} X^t$$

$$\Rightarrow \text{(ii): } U^t U = I_{(n-p) \times (n-p)}$$

Parameter Estimation (Cont'd)

REML

We first combine all models

$$Y_i \sim N(X_i \beta, V_i)$$

into one model

$$Y \sim N(X\beta, V)$$

in which

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_N \end{pmatrix}, X = \begin{pmatrix} X_1 \\ \vdots \\ X_N \end{pmatrix}, V(\alpha) = \begin{pmatrix} V_1 & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & V_N \end{pmatrix}$$

Parameter Estimation (Cont'd)

REML

Again, the data are transformed orthogonal to X

$$U = A^t Y \sim N(0, A^t V(\alpha) A)$$

The MLE of α , based on U is called the REML estimate, and is denoted by $\hat{\alpha}_{REML}$

The resulting estimate $\hat{\beta}(\hat{\alpha}_{REML})$ for β will be denoted by $\hat{\beta}_{REML}$

Parameter Estimation (Cont'd)

REML

Facts

$$(1) A(A^tVA)^{-1}A^t = V^{-1} - V^{-1}X(X^tV^{-1}X)^{-1}X^tV^{-1}$$
$$(2) |A^tVA|^{1/2} \propto |V|^{1/2} |X^tV^{-1}X|^{1/2}$$

Likelihood

$$A^tY \sim N(0, A^tV(\alpha)A),$$

$$\because A^tX\beta = A^tAA^tX\beta = A^t(I - X(X^tX)^{-1}X^t)X\beta = 0$$

$$\begin{aligned} l(\alpha; A^tY) &= -\frac{1}{2} \ln |A^tVA| - \frac{1}{2} Y^t A(A^tVA)^{-1} A^tY \\ &\propto -\frac{1}{2} \ln |X^tV^{-1}X| - \frac{1}{2} \ln |V| - \frac{1}{2} Y^t \left[V^{-1} - V^{-1}X(X^tV^{-1}X)^{-1}X^tV^{-1} \right] Y \\ &= -\frac{1}{2} \ln |X^tV^{-1}X| - \frac{1}{2} \ln |V| - \frac{1}{2} (Y - X\hat{\beta})^t V^{-1} (Y - X\hat{\beta}) \\ &= -\frac{1}{2} \ln |X^tV^{-1}X| + l_p(Y; \theta) \end{aligned}$$

**Profile
Likelihood**

Parameter Estimation (Cont'd)

REML

Proof of fact (1) $A(A'VA)^{-1}A' = V^{-1} - V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1}$

$$\begin{aligned} T &= \begin{pmatrix} X'V^{-1} \\ A' \end{pmatrix} \Rightarrow TVT' = \begin{pmatrix} X'V^{-1}X & X'A' \\ A'X & A'VA \end{pmatrix} = \begin{pmatrix} X'V^{-1}X & 0 \\ 0 & A'VA \end{pmatrix} \\ \Rightarrow (TVT')^{-1} &= \begin{pmatrix} (X'V^{-1}X)^{-1} & 0 \\ 0 & (A'VA)^{-1} \end{pmatrix} \\ \Rightarrow V^{-1} &= T' \begin{pmatrix} (X'V^{-1}X)^{-1} & 0 \\ 0 & (A'VA)^{-1} \end{pmatrix} T = V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1} + A(A'VA)^{-1}A' \\ \Rightarrow A(A'VA)^{-1}A' &= V^{-1} - V^{-1}X(X'V^{-1}X)^{-1}X'V^{-1} \end{aligned}$$

Parameter Estimation (Cont'd)

REML

Proof of fact (2) $|A^tVA|^{1/2} \propto |V|^{1/2} |X^tV^{-1}X|^{1/2}$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & I \end{pmatrix} \begin{pmatrix} I & A^{-1}B \\ 0 & D - CA^{-1}B \end{pmatrix} \\ = \begin{pmatrix} I & B \\ 0 & D \end{pmatrix} \begin{pmatrix} A - BD^{-1}C & 0 \\ D^{-1}C & I \end{pmatrix}$$

$$TY \equiv \begin{pmatrix} T_1 \\ T_2 \end{pmatrix} Y \equiv \begin{pmatrix} (X^tV^{-1}X)^{-1}X^tV^{-1} \\ A^t \end{pmatrix} Y \sim N \left\{ \begin{pmatrix} \beta \\ 0 \end{pmatrix}, \begin{pmatrix} (X^tV^{-1}X)^{-1} & 0 \\ 0 & A^tVA \end{pmatrix} \right\}$$

$$|V| \cdot |TT^t| = |T^tVT| = |A^tVA| \cdot |(X^tV^{-1}X)^{-1}|$$

If we let $G = (X^tV^{-1}X)^{-1}X^tV^{-1}$, then we have

$$\begin{aligned} |TT^t| &= \begin{vmatrix} GG^t & GA \\ A^tG^t & A^tA \end{vmatrix} = |A^tA| \cdot |GG^t - GA(A^tA)^{-1}A^tG^t| \\ &= |G(I - AA^t)G^t| \quad (\because A^tA = I) \\ &= |GX(X^tX)^{-1}X^tG^t| \quad (\because AA^t = I - X(X^tX)^{-1}X^t) \\ &= |X^tX|^{-1} \quad (\because G = (X^tV^{-1}X)^{-1}X^tV^{-1}) \end{aligned}$$

$$\text{Thus, } |V| \cdot |X^tX|^{-1} = |A^tVA| \cdot |(X^tV^{-1}X)^{-1}|$$

$$\Rightarrow |A^tVA| \propto |V| \cdot |X^tV^{-1}X|$$

Parameter Estimation (Cont'd)

Restricted likelihood

$\hat{\alpha}_{REML}$ and $\hat{\beta}_{REML}$ can also be obtained from maximizing

$$L_{REL}(\theta) = |X^t V(\alpha)^{-1} X|^{-\frac{1}{2}} L_{ML}(\theta)$$

$$l_{REL}(\theta) = -\frac{1}{2} \ln |X^t V(\alpha)^{-1} X| - \frac{1}{2} \ln |V(\alpha)| - \frac{1}{2} (Y - X\beta)^t V(\alpha)^{-1} (Y - X\beta)$$

with respect to $\theta = (\beta^t, \alpha^t)^t$, with respect to α and β simultaneously

Although strictly speaking no likelihood, $L_{REL}(\theta)$ is therefore still called the restricted likelihood function

Analysis with LMM

In practice: often unbalanced data

Unequal number of measurements per subject

Measurements not taken at fixed time points

Therefore, multivariate regression techniques such as MANOVA/ANOVA are often not applicable

Analysis with LMM (Cont'd)

Marginal model formulation:

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \boldsymbol{\varepsilon}_i, \boldsymbol{\varepsilon}_i \sim MVN(\mathbf{0}, \mathbf{R}_i)$$

Variance structure should be carefully selected

The marginal mean is obtained by integrating out individuals' heterogeneities and indicates a population average if and only if the individuals in the study can be regarded as a random sample from a population

Random effect model formulation:

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{b}_i + \boldsymbol{\varepsilon}_i, \mathbf{b}_i \sim MVN(\mathbf{0}, \mathbf{G}_i), \boldsymbol{\varepsilon}_i \sim MVN(\mathbf{0}, \mathbf{R}_i)$$

The difference between the conditional mean and the marginal mean of an individual is the random effect

$$\mathbf{b}_i = E(\mathbf{Y}_i | \mathbf{b}_i) - E(\mathbf{Y}_i)$$

Analysis with LMM (Cont'd)

Remarks

For linear mixed model, the coefficients for marginal model and random effect models are equivalent. However, they are usually different for non-normal distributions

When coefficients are different, random effect models are often preferred because we can have a simple marginal interpretation using the conditional model not only in normal random-effect models, but also in a wider class

Analysis with LMM (Cont'd)

Remarks

Marginal and random effect models for normally distributed data are equivalent except

Marginal model: $\mathbf{R}$ - positive definite

Marginal model: $\mathbf{G}$ & $\mathbf{R}$ - positive definite

$$\mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{Z}_i\mathbf{b}_i + \boldsymbol{\varepsilon}_i, \mathbf{b}_i \sim MVN(0, \mathbf{G}_i), \boldsymbol{\varepsilon}_i \sim MVN(0, \mathbf{R}_i)$$

$$\Rightarrow \mathbf{Y}_i = \mathbf{X}_i\boldsymbol{\beta} + \boldsymbol{\varepsilon}_i, \boldsymbol{\varepsilon}_i \sim MVN(0, \mathbf{Z}_i\mathbf{G}_i\mathbf{Z}_i^t + \mathbf{R}_i)$$

Random effect model is sometimes called conditional model or subject-based model

Marginal Model

Model formulation

Identify the mean model

$$\mathbf{Y}_i = \mathbf{X}_i \boldsymbol{\beta} + \boldsymbol{\varepsilon}_i, \boldsymbol{\varepsilon}_i \sim MVN(\mathbf{0}, \mathbf{R}_i)$$

Select the variance-covariance matrix structure

Confirm the mean model

Random Effect Model

Model formulation

Two stage model

Stage 1: Linear regression model for each subject separately

Stage 2: Explain variability in the subject-specific regression coefficients using known covariances

Notation

Response Y_{ij} for i th subject, measured at time t_{ij} , $i = 1, \dots, N$, $j = 1, \dots, n_i$

Response vector Y_i for i th subject:

$$Y_i = (Y_{i1}, Y_{i2}, \dots, Y_{in_i})'$$

Random Effect Model (Cont'd)

Stage 1

Model

$$\mathbf{Y}_i = \mathbf{Z}_i \boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_i,$$

$\mathbf{Z}_i$ is a $(n_i \times q)$ matrix of known covariates

$\boldsymbol{\beta}_i$ is a q -dimensional vector of subject-specific regression coefficients

$\boldsymbol{\varepsilon}_i \sim N(0, R_i)$, often $R_i = \sigma^2 I_{n_i}$

Note that the above model describes the observed variability with subjects

Random Effect Model (Cont'd)

Stage 2

Between-subject variability can now be studied from relating β_i to known covariates

Stage 2 model:

$$\beta_i = K_i \beta + b_i$$

K_i is a $(q \times p)$ matrix of known covariates

β_i is a p -dimensional vector of unknown regression parameters

$$b_i \sim N(0, G_i)$$

Random Effect Model (Cont'd)

Graphical representation

